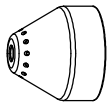


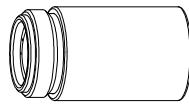
## TORCH SETUP

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### 85 A shielded consumables



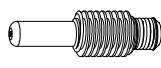
220817  
Shield



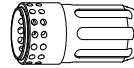
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Retaining cap



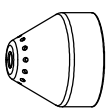
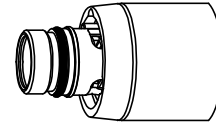
220816  
Nozzle



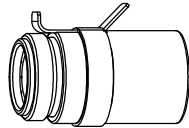
220842  
Electrode



220857  
Swirl ring



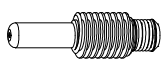
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Shield



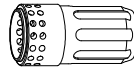
220953  
Ohmic-sensing  
retaining cap



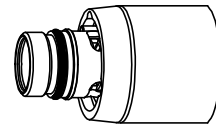
220816  
Nozzle



220842  
Electrode



220857  
Swirl ring



## TORCH SETUP

**85A Shielded**

**Mild Steel**

| Air flow rate - slpm/scfh |           |
|---------------------------|-----------|
| Hot                       | 190 / 400 |
| Cold                      | 235 / 500 |

**Metric**

| Material Thickness | Torch-to-Work Distance | Initial Pierce Height |     | Pierce Delay Time | Recommended |         | Maximum   |         |     |
|--------------------|------------------------|-----------------------|-----|-------------------|-------------|---------|-----------|---------|-----|
|                    |                        |                       |     |                   | Cut Speed   | Voltage | Cut Speed | Voltage |     |
| mm                 | mm                     | mm                    | %   | seconds           | (mm/min)    | Volts   | (mm/min)  | Volts   |     |
| 3                  | 1.5                    | 3.8                   | 250 | 0.1               | 6800        | 122     | 11500     | 115     |     |
| 4                  |                        |                       |     | 0.2               | 5650        | 122     | 9100      | 119     |     |
| 6                  |                        |                       |     | 0.5               | 3600        | 123     | 5500      | 126     |     |
| 8                  |                        |                       |     |                   | 2500        | 125     | 3900      | 129     |     |
| 10                 |                        |                       |     |                   | 1680        | 127     | 2600      | 129     |     |
| 12                 |                        | 4.5                   | 300 | 0.7               | 1280        | 130     | 2000      | 130     |     |
| 16                 |                        |                       |     | 1.0               | 870         | 134     | 1150      | 132     |     |
| 20                 |                        |                       |     | 1.5               | 570         | 137     | 850       | 135     |     |
| 25                 |                        | Edge Start            |     |                   |             | 350     | 142       | 550     | 139 |
| 30                 |                        |                       |     |                   |             | 200     | 146       | 370     | 143 |

**English**

| Material Thickness | Torch-to-Work Distance | Initial Pierce Height |     | Pierce Delay Time | Recommended |         | Maximum   |         |
|--------------------|------------------------|-----------------------|-----|-------------------|-------------|---------|-----------|---------|
|                    |                        |                       |     |                   | Cut Speed   | Voltage | Cut Speed | Voltage |
|                    | in                     | in                    | %   | seconds           | ipm         | Volts   | ipm       | Volts   |
| 10GA               | 0.06                   | 0.15                  | 250 | 0.0               | 250         | 122     | 420       | 117     |
| 3/16 in            |                        |                       |     | 0.2               | 185         | 123     | 275       | 121     |
| 1/4 in             |                        |                       |     | 0.5               | 130         | 123     | 200       | 128     |
| 3/8 in             |                        |                       |     |                   | 70          | 126     | 108       | 129     |
| 1/2 in             |                        | 0.18                  | 300 | 45                | 131         | 70      | 130       |         |
| 5/8 in             |                        |                       |     | 1.0               | 35          | 134     | 46        | 132     |
| 3/4 in             |                        |                       |     | 1.5               | 24          | 136     | 36        | 134     |
| 7/8 in             |                        | Edge Start            |     |                   | 19          | 139     | 27        | 136     |
| 1 in               |                        |                       |     |                   | 13          | 142     | 21        | 140     |
| 1-1/8 in           |                        |                       |     |                   | 9           | 145     | 16        | 142     |
| 1-1/4 in           |                        |                       |     |                   | 7           | 148     | 13        | 145     |

## TORCH SETUP

**85A Shielded  
Stainless Steel**

| Air flow rate - slpm/scfh |           |
|---------------------------|-----------|
| Hot                       | 190 / 400 |
| Cold                      | 235 / 500 |

### Metric

| Material Thickness | Torch-to-Work Distance | Initial Pierce Height |     | Pierce Delay Time | Recommended |         | Maximum   |         |     |
|--------------------|------------------------|-----------------------|-----|-------------------|-------------|---------|-----------|---------|-----|
|                    |                        |                       |     |                   | Cut Speed   | Voltage | Cut Speed | Voltage |     |
| mm                 | mm                     | mm                    | %   | seconds           | (mm/min)    | Volts   | (mm/min)  | Volts   |     |
| 3                  | 1.5                    | 3.8                   | 250 | 0.1               | 7500        | 122     | 11500     | 115     |     |
| 4                  |                        |                       |     | 0.2               | 6100        | 122     | 9400      | 117     |     |
| 6                  |                        |                       |     | 0.5               | 3700        | 122     | 5700      | 121     |     |
| 8                  |                        |                       |     |                   | 2450        | 124     | 3800      | 123     |     |
| 10                 |                        |                       |     |                   | 1550        | 127     | 2400      | 124     |     |
| 12                 |                        | 4.5                   | 300 | 0.7               | 1100        | 131     | 1750      | 125     |     |
| 16                 |                        |                       |     | 1.0               | 700         | 135     | 950       | 131     |     |
| 20                 |                        | Edge Start            |     |                   |             | 480     | 138       | 700     | 132 |
| 25                 |                        |                       |     |                   |             | 300     | 143       | 480     | 135 |

### English

| Material Thickness | Torch-to-Work Distance | Initial Pierce Height |     | Pierce Delay Time | Recommended |         | Maximum   |         |
|--------------------|------------------------|-----------------------|-----|-------------------|-------------|---------|-----------|---------|
|                    |                        |                       |     |                   | Cut Speed   | Voltage | Cut Speed | Voltage |
|                    | in                     | in                    | %   | seconds           | ipm         | Volts   | ipm       | Volts   |
| 10GA               | 0.06                   | 0.15                  | 250 | 0.2               | 275         | 122     | 420       | 116     |
| 3/16 in            |                        |                       |     |                   | 200         | 122     | 300       | 119     |
| 1/4 in             |                        |                       |     | 0.5               | 130         | 122     | 205       | 122     |
| 3/8 in             |                        |                       |     |                   | 65          | 126     | 100       | 124     |
| 1/2 in             |                        |                       |     |                   | 36          | 132     | 60        | 125     |
| 5/8 in             |                        | 0.18                  | 300 | 1.0               | 28          | 135     | 38        | 131     |
| 3/4 in             |                        |                       |     | Edge Start        |             |         |           |         |
| 7/8 in             |                        |                       |     |                   |             |         |           |         |
| 1 in               |                        |                       |     |                   |             |         |           |         |

## TORCH SETUP

### 85A Shielded

### Aluminum

#### Air flow rate - slpm/scfh

|     |           |
|-----|-----------|
| Hot | 190 / 400 |
|-----|-----------|

|      |           |
|------|-----------|
| Cold | 235 / 500 |
|------|-----------|

### Metric

| Material Thickness | Torch-to-Work Distance | Initial Pierce Height |     | Pierce Delay Time | Recommended |         | Maximum   |         |     |
|--------------------|------------------------|-----------------------|-----|-------------------|-------------|---------|-----------|---------|-----|
|                    |                        |                       |     |                   | Cut Speed   | Voltage | Cut Speed | Voltage |     |
| mm                 | mm                     | mm                    | %   | seconds           | (mm/min)    | Volts   | (mm/min)  | Volts   |     |
| 3                  | 1.5                    | 3.8                   | 250 | 0.1               | 8000        | 122     | 11800     | 117     |     |
| 4                  |                        |                       |     | 0.2               | 6500        | 123     | 10000     | 118     |     |
| 6                  |                        |                       |     | 0.5               | 3800        | 126     | 6100      | 121     |     |
| 8                  |                        |                       |     |                   | 2650        | 130     | 4300      | 123     |     |
| 10                 |                        |                       |     |                   | 1920        | 132     | 3100      | 125     |     |
| 12                 |                        | 4,5                   | 300 | 0.7               | 1450        | 134     | 2400      | 127     |     |
| 16                 |                        |                       |     | 1.0               | 950         | 139     | 1500      | 130     |     |
| 20                 |                        | Edge Start            |     |                   |             | 600     | 143       | 1100    | 133 |
| 25                 |                        |                       |     |                   |             | 380     | 146       | 670     | 140 |

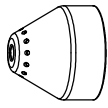
### English

| Material Thickness | Torch-to-Work Distance | Initial Pierce Height |     | Pierce Delay Time | Recommended |         | Maximum   |         |
|--------------------|------------------------|-----------------------|-----|-------------------|-------------|---------|-----------|---------|
|                    |                        |                       |     |                   | Cut Speed   | Voltage | Cut Speed | Voltage |
|                    | in                     | in                    | %   | seconds           | ipm         | Volts   | ipm       | Volts   |
| 1/8 in             | 0.06                   | 0.15                  | 250 | 0.2               | 300         | 122     | 450       | 117     |
| 1/4 in             |                        |                       |     | 0.5               | 130         | 127     | 215       | 122     |
| 3/8 in             |                        |                       |     |                   | 80          | 132     | 130       | 124     |
| 1/2 in             |                        | 0.18                  | 300 | 50                | 135         | 85      | 128       |         |
| 5/8 in             |                        |                       |     | 38                | 139         | 60      | 130       |         |
| 3/4 in             |                        |                       |     | Edge Start        |             |         | 25        | 142     |
| 7/8 in             |                        | 20                    | 144 |                   |             |         | 36        | 137     |
| 1 in               |                        | 14                    | 146 |                   |             |         | 25        | 141     |

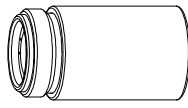
## TORCH SETUP

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### 65 A shielded consumables



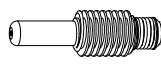
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Shield



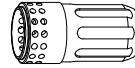
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Retaining cap



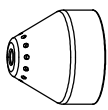
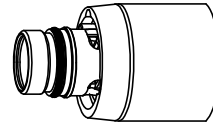
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Nozzle



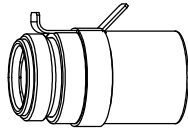
220842  
Electrode



220857  
Swirl ring



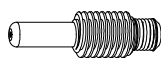
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Shield



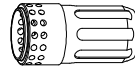
220953  
Ohmic-sensing  
retaining cap



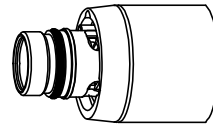
220819  
Nozzle



220842  
Electrode



220857  
Swirl ring



## TORCH SETUP

65A Shielded

Mild Steel

Air flow rate - slpm/scfh

Hot 160 / 340

Cold 220 / 470

Metric

| Material Thickness | Torch-to-Work Distance | Initial Pierce Height |     | Pierce Delay Time | Recommended |         | Maximum   |         |     |
|--------------------|------------------------|-----------------------|-----|-------------------|-------------|---------|-----------|---------|-----|
|                    |                        |                       |     |                   | Cut Speed   | Voltage | Cut Speed | Voltage |     |
| mm                 | mm                     | mm                    | %   | seconds           | (mm/min)    | Volts   | (mm/min)  | Volts   |     |
| 2                  | 1.5                    | 3.8                   | 250 | 0.1               | 6050        | 124     | 8800      | 120     |     |
| 3                  |                        |                       |     | 0.2               | 5200        | 125     | 7600      | 122     |     |
| 4                  |                        |                       |     | 0.5               | 4250        | 125     | 6400      | 123     |     |
| 6                  |                        |                       |     |                   | 2550        | 127     | 4000      | 129     |     |
| 8                  |                        |                       |     |                   | 1700        | 129     | 2800      | 130     |     |
| 10                 |                        | 4.5                   | 300 | 0.7               | 1100        | 131     | 1900      | 130     |     |
| 12                 |                        |                       |     | 1.2               | 850         | 134     | 1400      | 132     |     |
| 16                 |                        |                       |     | 2.0               | 560         | 138     | 800       | 136     |     |
| 20                 |                        | Edge Start            |     |                   |             | 350     | 142       | 560     | 140 |
| 25                 |                        |                       |     |                   |             | 210     | 145       | 320     | 143 |

English

| Material Thickness | Torch-to-Work Distance | Initial Pierce Height |     | Pierce Delay Time | Recommended |         | Maximum   |         |
|--------------------|------------------------|-----------------------|-----|-------------------|-------------|---------|-----------|---------|
|                    |                        |                       |     |                   | Cut Speed   | Voltage | Cut Speed | Voltage |
|                    | in                     | in                    | %   | seconds           | ipm         | Volts   | ipm       | Volts   |
| 16GA               | 0.06                   | 0.15                  | 250 | 0.1               | 260         | 123     | 370       | 118     |
| 10GA               |                        |                       |     |                   | 190         | 125     | 280       | 123     |
| 3/16 in            |                        |                       |     | 0.2               | 140         | 126     | 210       | 124     |
| 1/4 in             |                        |                       |     | 0.5               | 90          | 127     | 145       | 130     |
| 3/8 in             |                        | 0.18                  | 300 | 0.7               | 45          | 130     | 78        | 130     |
| 1/2 in             |                        |                       |     | 1.2               | 30          | 135     | 50        | 132     |
| 5/8 in             |                        |                       |     | 2.0               | 23          | 138     | 32        | 136     |
| 3/4 in             |                        |                       |     | Edge Start        |             |         | 15        | 141     |
| 7/8 in             |                        | 12                    | 143 |                   |             |         | 18        | 141     |
| 1 in               |                        | 8                     | 145 |                   |             |         | 12        | 143     |

## TORCH SETUP

**65A Shielded  
Stainless Steel**

| Air flow rate - slpm/scfh |           |
|---------------------------|-----------|
| Hot                       | 160 / 340 |
| Cold                      | 220 / 470 |

**Metric**

| Material Thickness | Torch-to-Work Distance | Initial Pierce Height |     | Pierce Delay Time | Recommended |         | Maximum   |         |     |
|--------------------|------------------------|-----------------------|-----|-------------------|-------------|---------|-----------|---------|-----|
|                    |                        |                       |     |                   | Cut Speed   | Voltage | Cut Speed | Voltage |     |
| mm                 | mm                     | mm                    | %   | seconds           | (mm/min)    | Volts   | (mm/min)  | Volts   |     |
| 2                  | 1.5                    | 3.8                   | 250 | 0.1               | 8100        | 125     | 12500     | 122     |     |
| 3                  |                        |                       |     | 0.2               | 6700        | 125     | 10500     | 123     |     |
| 4                  |                        |                       |     | 0.5               | 5200        | 125     | 7700      | 124     |     |
| 6                  |                        |                       |     |                   | 2450        | 126     | 3600      | 126     |     |
| 8                  |                        |                       |     | 0.7               | 1500        | 129     | 2300      | 127     |     |
| 10                 |                        | 960                   | 132 |                   | 1550        | 127     |           |         |     |
| 12                 |                        | 4.5                   | 300 | 1.2               | 750         | 135     | 1150      | 129     |     |
| 16                 |                        | Edge Start            |     |                   |             | 500     | 139       | 650     | 134 |
| 20                 |                        |                       |     |                   |             | 300     | 143       | 450     | 136 |

**English**

| Material Thickness | Torch-to-Work Distance | Initial Pierce Height |     | Pierce Delay Time | Recommended |         | Maximum   |         |     |
|--------------------|------------------------|-----------------------|-----|-------------------|-------------|---------|-----------|---------|-----|
|                    |                        |                       |     |                   | Cut Speed   | Voltage | Cut Speed | Voltage |     |
|                    | in                     | in                    | %   | seconds           | ipm         | Volts   | ipm       | Volts   |     |
| 16GA               | 0.06                   | 0.15                  | 250 | 0.1               | 345         | 124     | 530       | 121     |     |
| 10GA               |                        |                       |     |                   | 240         | 125     | 370       | 123     |     |
| 3/16 in            |                        |                       |     | 0.2               | 155         | 126     | 210       | 125     |     |
| 1/4 in             |                        |                       |     | 0.5               | 80          | 126     | 120       | 126     |     |
| 3/8 in             |                        | 0.18                  | 300 | 0.7               | 40          | 131     | 65        | 127     |     |
| 1/2 in             |                        |                       |     | 1.2               | 26          | 136     | 40        | 129     |     |
| 5/8 in             |                        | Edge Start            |     |                   |             | 20      | 139       | 25      | 134 |
| 3/4 in             |                        |                       |     |                   |             | 14      | 142       | 19      | 136 |

## TORCH SETUP

### 65A Shielded

### Aluminum

| Air flow rate - slpm/scfh |           |
|---------------------------|-----------|
| Hot                       | 160 / 340 |
| Cold                      | 220 / 470 |

### Metric

| Material Thickness | Torch-to-Work Distance | Initial Pierce Height |     | Pierce Delay Time | Recommended |         | Maximum   |         |
|--------------------|------------------------|-----------------------|-----|-------------------|-------------|---------|-----------|---------|
|                    |                        |                       |     |                   | Cut Speed   | Voltage | Cut Speed | Voltage |
| mm                 | mm                     | mm                    | %   | seconds           | (mm/min)    | Volts   | (mm/min)  | Volts   |
| 2                  | 1.5                    | 3.8                   | 250 | 0.1               | 8800        | 121     | 12800     | 122     |
| 3                  |                        |                       |     | 0.2               | 7400        | 124     | 11000     | 123     |
| 4                  |                        |                       |     | 0.5               | 6000        | 126     | 9150      | 124     |
| 6                  |                        |                       |     |                   | 3200        | 130     | 5500      | 127     |
| 8                  |                        |                       |     | 0.7               | 1950        | 133     | 3450      | 129     |
| 10                 |                        | 4.5                   | 300 |                   | 1200        | 136     | 2050      | 130     |
| 12                 |                        |                       | 1.2 | 1000              | 138         | 1650    | 132       |         |
| 16                 |                        | Edge Start            |     |                   | 650         | 143     | 1000      | 134     |
| 20                 |                        |                       |     |                   | 380         | 147     | 700       | 137     |

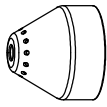
### English

| Material Thickness | Torch-to-Work Distance | Initial Pierce Height |     | Pierce Delay Time | Recommended |         | Maximum   |         |
|--------------------|------------------------|-----------------------|-----|-------------------|-------------|---------|-----------|---------|
|                    |                        |                       |     |                   | Cut Speed   | Voltage | Cut Speed | Voltage |
|                    | in                     | in                    | %   | seconds           | ipm         | Volts   | ipm       | Volts   |
| 1/16 in            | 0.06                   | 0.15                  | 250 | 0.1               | 365         | 121     | 535       | 121     |
| 1/8 in             |                        |                       |     |                   | 280         | 124     | 420       | 123     |
| 1/4 in             |                        |                       |     | 0.5               | 105         | 131     | 190       | 127     |
| 3/8 in             |                        | 0.18                  | 300 | 0.7               | 50          | 135     | 85        | 130     |
| 1/2 in             |                        |                       |     | 1.2               | 35          | 139     | 60        | 133     |
| 5/8 in             |                        |                       |     | Edge Start        |             |         | 26        | 143     |
| 3/4 in             |                        | Edge Start            |     |                   | 16          | 146     | 30        | 136     |

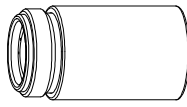
## TORCH SETUP

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### 45 A shielded consumables



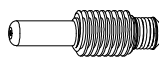
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Shield



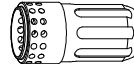
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Retaining cap



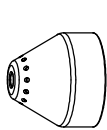
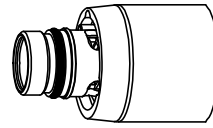
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Nozzle



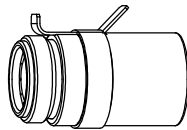
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Electrode



220857  
Swirl ring



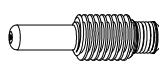
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Shield



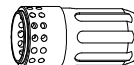
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Ohmic-sensing  
retaining cap



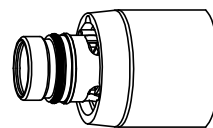
220941  
Nozzle



220842  
Electrode



220857  
Swirl ring



## TORCH SETUP

### 45A Shielded

### Mild Steel

| Air flow rate - slpm/scfh |           |
|---------------------------|-----------|
| Hot                       | 147 / 310 |
| Cold                      | 210/ 450  |

### Metric

| Material Thickness | Torch-to-Work Distance | Initial Pierce Height |     | Pierce Delay Time | Recommended |         | Maximum   |         |
|--------------------|------------------------|-----------------------|-----|-------------------|-------------|---------|-----------|---------|
|                    |                        |                       |     |                   | Cut Speed   | Voltage | Cut Speed | Voltage |
| mm                 | mm                     | mm                    | %   | seconds           | (mm/min)    | Volts   | (mm/min)  | Volts   |
| 0.5                | 1.5                    | 3.8                   | 250 | 0.0               | 9000        | 128     | 12700     | 128     |
| 1                  |                        |                       |     |                   | 9000        | 128     | 12700     | 128     |
| 1.5                |                        |                       |     | 0.1               | 9000        | 130     | 12700     | 128     |
| 2                  |                        |                       |     | 0.3               | 6600        | 130     | 9750      | 128     |
| 3                  |                        |                       |     | 0.4               | 3850        | 133     | 6150      | 130     |
| 4                  |                        |                       |     |                   | 2200        | 134     | 4450      | 131     |
| 6                  |                        |                       |     | 0.5               | 1350        | 137     | 2550      | 132     |

### English

| Material Thickness | Torch-to-Work Distance | Initial Pierce Height |     | Pierce Delay Time | Recommended |         | Maximum   |         |
|--------------------|------------------------|-----------------------|-----|-------------------|-------------|---------|-----------|---------|
|                    |                        |                       |     |                   | Cut Speed   | Voltage | Cut Speed | Voltage |
|                    | in                     | in                    | %   | seconds           | ipm         | Volts   | ipm       | Volts   |
| 26GA               | 0.06                   | 0.15                  | 250 | 0.0               | 350         | 128     | 500       | 128     |
| 22GA               |                        |                       |     |                   | 350         | 128     | 500       | 128     |
| 18GA               |                        |                       |     | 0.1               | 350         | 129     | 500       | 128     |
| 16GA               |                        |                       |     |                   | 350         | 130     | 500       | 128     |
| 14GA               |                        |                       |     | 0.2               | 270         | 130     | 400       | 128     |
| 12GA               |                        |                       |     | 0.4               | 190         | 133     | 270       | 130     |
| 10GA               |                        |                       |     |                   | 100         | 134     | 205       | 131     |
| 3/16 in            |                        |                       |     | 0.5               | 70          | 135     | 135       | 132     |
| 1/4 in             |                        |                       |     | 0.6               | 48          | 137     | 91        | 132     |

## TORCH SETUP

**45A Shielded  
Stainless Steel**

| Air flow rate - slpm/scfh |           |
|---------------------------|-----------|
| Hot                       | 147 / 310 |
| Cold                      | 210/ 450  |

**Metric**

| Material Thickness | Torch-to-Work Distance | Initial Pierce Height |     | Pierce Delay Time | Recommended |         | Maximum   |         |
|--------------------|------------------------|-----------------------|-----|-------------------|-------------|---------|-----------|---------|
|                    |                        |                       |     |                   | Cut Speed   | Voltage | Cut Speed | Voltage |
| mm                 | mm                     | mm                    | %   | seconds           | (mm/min)    | Volts   | (mm/min)  | Volts   |
| 0.5                | 1.5                    | 3.8                   | 250 | 0.0               | 9000        | 130     | 12700     | 130     |
| 1                  |                        |                       |     |                   | 9000        | 130     | 12700     | 130     |
| 1.5                |                        |                       |     | 0.1               | 9000        | 130     | 12700     | 130     |
| 2                  |                        |                       |     | 0.3               | 6000        | 132     | 10800     | 130     |
| 3                  |                        |                       |     | 0.4               | 3100        | 132     | 5550      | 131     |
| 4                  |                        |                       |     |                   | 2000        | 134     | 3250      | 132     |
| 6                  |                        |                       |     | 0.5               | 900         | 140     | 1250      | 138     |

**English**

| Material Thickness | Torch-to-Work Distance | Initial Pierce Height |     | Pierce Delay Time | Recommended |         | Maximum   |         |
|--------------------|------------------------|-----------------------|-----|-------------------|-------------|---------|-----------|---------|
|                    |                        |                       |     |                   | Cut Speed   | Voltage | Cut Speed | Voltage |
|                    | in                     | in                    | %   | seconds           | ipm         | Volts   | ipm       | Volts   |
| 26GA               | 0.06                   | 0.15                  | 250 | 0.0               | 350         | 130     | 500       | 130     |
| 22GA               |                        |                       |     |                   | 350         | 130     | 500       | 130     |
| 18GA               |                        |                       |     | 0.1               | 350         | 130     | 500       | 130     |
| 16GA               |                        |                       |     |                   | 350         | 130     | 500       | 130     |
| 14GA               |                        |                       |     | 0.2               | 250         | 132     | 450       | 130     |
| 12GA               |                        |                       |     | 0.4               | 140         | 132     | 258       | 130     |
| 10GA               |                        |                       |     |                   | 100         | 133     | 168       | 135     |
| 3/16 in            |                        |                       |     | 0.5               | 52          | 135     | 73        | 133     |
| 1/4 in             |                        |                       |     | 0.6               | 30          | 141     | 44        | 140     |

## TORCH SETUP

### 45A Shielded

### Aluminum

| Air flow rate - slpm/scfh |           |
|---------------------------|-----------|
| Hot                       | 147 / 310 |
| Cold                      | 210/ 450  |

### Metric

| Material Thickness | Torch-to-Work Distance | Initial Pierce Height |     | Pierce Delay Time | Recommended |         | Maximum   |         |
|--------------------|------------------------|-----------------------|-----|-------------------|-------------|---------|-----------|---------|
|                    |                        |                       |     |                   | Cut Speed   | Voltage | Cut Speed | Voltage |
| mm                 | mm                     | mm                    | %   | seconds           | (mm/min)    | Volts   | (mm/min)  | Volts   |
| 1                  | 1.5                    | 3.8                   | 250 | 0.0               | 8250        | 136     | 12700     | 136     |
| 2                  |                        |                       |     | 0.1               | 6600        | 136     | 11500     | 133     |
| 3                  |                        |                       |     | 0.2               | 3100        | 139     | 7800      | 133     |
| 4                  |                        |                       |     | 0.4               | 2200        | 141     | 6050      | 134     |
| 6                  |                        |                       |     | 0.5               | 1500        | 142     | 3500      | 136     |

### English

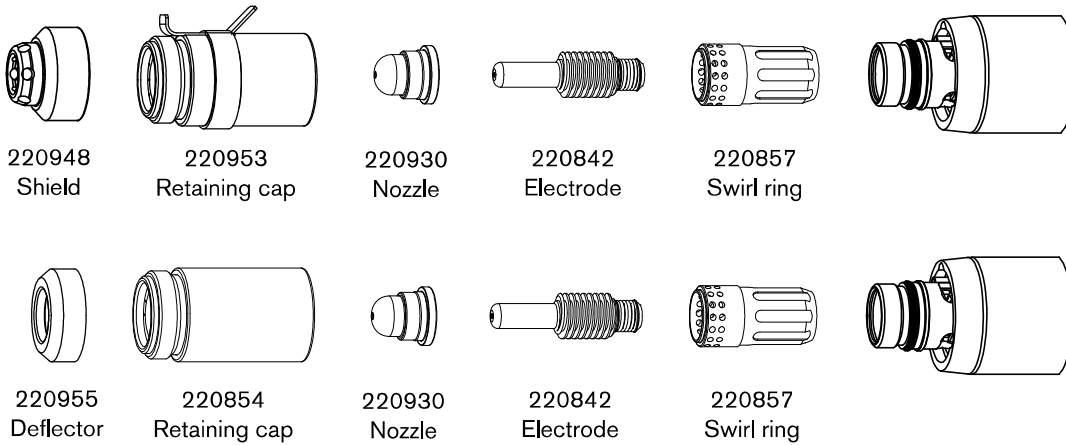
| Material Thickness | Torch-to-Work Distance | Initial Pierce Height |     | Pierce Delay Time | Recommended |         | Maximum   |         |
|--------------------|------------------------|-----------------------|-----|-------------------|-------------|---------|-----------|---------|
|                    |                        |                       |     |                   | Cut Speed   | Voltage | Cut Speed | Voltage |
|                    | in                     | in                    | %   | seconds           | ipm         | Volts   | ipm       | Volts   |
| 1/32 in            | 0.06                   | 0.15                  | 250 | 0.0               | 325         | 136     | 500       | 137     |
| 1/16 in            |                        |                       |     | 0.1               | 325         | 136     | 500       | 135     |
| 3/32 in            |                        |                       |     | 0.2               | 200         | 136     | 410       | 132     |
| 1/8 in             |                        |                       |     | 0.4               | 100         | 140     | 280       | 133     |
| 1/4 in             |                        |                       |     | 0.5               | 54          | 142     | 120       | 136     |

## TORCH SETUP

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### FineCut® consumables

Note: The cut charts in this section apply to both shielded and unshielded consumables



## TORCH SETUP

**FineCut**  
**Mild Steel**

| Air flow rate - slpm/scfh |           |
|---------------------------|-----------|
| Hot                       | 155 / 330 |
| Cold                      | 215 / 460 |

**Metric**

| Material Thickness | Amps | Torch-to-Work Distance | Initial Pierce Height |     | Pierce Delay Time | Recommended |         | Maximum   |         |
|--------------------|------|------------------------|-----------------------|-----|-------------------|-------------|---------|-----------|---------|
|                    |      |                        |                       |     |                   | Cut Speed   | Voltage | Cut Speed | Voltage |
| mm                 | A    | mm                     | mm                    | %   | seconds           | (mm/min)    | Volts   | (mm/min)  | Volts   |
| 0.5                | 40   | 1.5                    | 3.8                   | 250 | 0                 | 8250        | 78      | 12700     | 80      |
| 0.6                |      |                        |                       |     |                   | 8250        | 78      | 12700     | 81      |
| 0.8                |      |                        |                       |     | 0.1               | 8250        | 78      | 12700     | 80      |
| 1                  | 45   |                        |                       |     | 0.2               | 8250        | 78      | 12700     | 82      |
| 1.5                |      |                        |                       |     | 0.4               | 6400        | 78      | 8500      | 81      |
| 2                  |      |                        |                       |     |                   | 5250        | 82      | 6250      | 81      |
| 3                  |      |                        |                       |     | 0.5               | 2750        | 83      | 3650      | 81      |
| 4                  |      |                        |                       |     | 0.6               | 1900        | 84      | 2450      | 81      |

**English**

| Material Thickness | Amps | Torch-to-Work Distance | Initial Pierce Height |     | Pierce Delay Time | Recommended |         | Maximum   |         |
|--------------------|------|------------------------|-----------------------|-----|-------------------|-------------|---------|-----------|---------|
|                    |      |                        |                       |     |                   | Cut Speed   | Voltage | Cut Speed | Voltage |
|                    | A    | in                     | in                    | %   | seconds           | ipm         | Volts   | ipm       | Volts   |
| 26GA               | 40   | 0.06                   | 0.15                  | 250 | 0.0               | 325         | 78      | 500       | 80      |
| 24GA               |      |                        |                       |     |                   | 325         | 78      | 500       | 81      |
| 22GA               |      |                        |                       |     | 0.1               | 325         | 78      | 500       | 80      |
| 20GA               |      |                        |                       |     |                   | 325         | 78      | 500       | 82      |
| 18GA               | 45   |                        |                       |     | 0.2               | 325         | 78      | 440       | 81      |
| 16GA               |      |                        |                       |     | 0.4               | 250         | 78      | 330       | 81      |
| 14GA               |      |                        |                       |     |                   | 220         | 82      | 260       | 81      |
| 12GA               |      |                        |                       |     | 0.5               | 120         | 83      | 160       | 81      |
| 10GA               |      |                        |                       |     |                   | 95          | 84      | 124       | 81      |

## TORCH SETUP

**FineCut**  
**Stainless Steel**

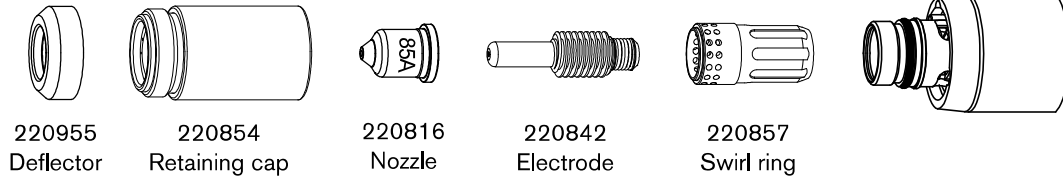
| Air flow rate - slpm/scfh |           |
|---------------------------|-----------|
| Hot                       | 155 / 330 |
| Cold                      | 215 / 460 |

**Metric**

| Material Thickness | Amps | Torch-to-Work Distance | Initial Pierce Height |     | Pierce Delay Time | Recommended |         | Maximum   |         |
|--------------------|------|------------------------|-----------------------|-----|-------------------|-------------|---------|-----------|---------|
|                    |      |                        |                       |     |                   | Cut Speed   | Voltage | Cut Speed | Voltage |
| mm                 | A    | mm                     | mm                    | %   | seconds           | (mm/min)    | Volts   | (mm/min)  | Volts   |
| 0.5                | 40   | 0.5                    | 2.0                   | 400 | 0                 | 8250        | 68      | 12700     | 63      |
| 0.6                |      |                        |                       |     |                   | 8250        | 68      | 12700     | 65      |
| 0.8                |      |                        |                       |     | 0.1               | 8250        | 68      | 12700     | 64      |
| 1                  | 45   |                        |                       |     | 0.15              | 8250        | 68      | 12700     | 65      |
| 1.5                |      |                        |                       |     | 0.4               | 6150        | 70      | 10650     | 64      |
| 2                  |      |                        |                       |     |                   | 4800        | 71      | 8150      | 64      |
| 3                  |      |                        |                       |     | 0.5               | 2550        | 81      | 3250      | 68      |
| 4                  |      |                        |                       |     | 0.6               | 1050        | 84      | 1250      | 72      |

**English**

| Material Thickness | Amps | Torch-to-Work Distance | Initial Pierce Height |     | Pierce Delay Time | Recommended |         | Maximum   |         |
|--------------------|------|------------------------|-----------------------|-----|-------------------|-------------|---------|-----------|---------|
|                    |      |                        |                       |     |                   | Cut Speed   | Voltage | Cut Speed | Voltage |
|                    | A    | in                     | in                    | %   | seconds           | ipm         | Volts   | ipm       | Volts   |
| 26GA               | 40   | 0.02                   | 0.08                  | 400 | 0.0               | 325         | 68      | 500       | 68      |
| 24GA               |      |                        |                       |     |                   | 325         | 68      | 500       | 65      |
| 22GA               |      |                        |                       |     | 0.1               | 325         | 68      | 500       | 64      |
| 20GA               |      |                        |                       |     |                   | 325         | 68      | 500       | 65      |
| 18GA               | 45   |                        |                       |     | 0.2               | 325         | 68      | 500       | 65      |
| 16GA               |      |                        |                       |     | 0.4               | 240         | 70      | 410       | 64      |
| 14GA               |      |                        |                       |     |                   | 200         | 70      | 345       | 64      |
| 12GA               |      |                        |                       |     | 0.5               | 120         | 80      | 155       | 67      |
| 10GA               |      |                        |                       |     | 0.6               | 75          | 83      | 95        | 70      |

**85 A unshielded consumables**

## TORCH SETUP

85A Unshielded

Mild Steel

| Air flow rate - slpm/scfh |           |
|---------------------------|-----------|
| Hot                       | 190 / 400 |
| Cold                      | 235 / 500 |

Metric

| Material Thickness | Torch-to-Work Distance | Initial Pierce Height |     | Pierce Delay Time | Recommended |         | Maximum   |         |
|--------------------|------------------------|-----------------------|-----|-------------------|-------------|---------|-----------|---------|
|                    |                        |                       |     |                   | Cut Speed   | Voltage | Cut Speed | Voltage |
| mm                 | mm                     | mm                    | %   | seconds           | (mm/min)    | Volts   | (mm/min)  | Volts   |
| 2                  | 2.0                    | 5.0                   | 250 | 0.0               | 7150        | 117     | 13000     | 113     |
| 3                  |                        |                       |     | 0.1               | 6240        | 118     | 11200     | 114     |
| 4                  |                        |                       |     | 0.2               | 5250        | 118     | 9000      | 117     |
| 6                  |                        |                       |     | 0.5               | 3450        | 120     | 5500      | 120     |
| 8                  |                        |                       |     |                   | 2400        | 121     | 3900      | 121     |
| 10                 |                        | 6.0                   | 300 | 0.7               | 1560        | 123     | 2600      | 122     |
| 12                 |                        |                       |     |                   | 1200        | 126     | 2000      | 124     |
| 16                 |                        |                       |     |                   | 820         | 132     | 1150      | 126     |
| 20                 |                        |                       |     |                   | 540         | 137     | 800       | 131     |
| 25                 |                        |                       |     |                   | 320         | 143     | 500       | 137     |

English

| Material Thickness | Torch-to-Work Distance | Initial Pierce Height |     | Pierce Delay Time | Recommended |         | Maximum   |         |
|--------------------|------------------------|-----------------------|-----|-------------------|-------------|---------|-----------|---------|
|                    |                        |                       |     |                   | Cut Speed   | Voltage | Cut Speed | Voltage |
|                    | in                     | in                    | %   | seconds           | ipm         | Volts   | ipm       | Volts   |
| 14GA               | 0.08                   | 0.20                  | 250 | 0.1               | 300         | 116     | 540       | 112     |
| 10GA               |                        |                       |     | 0.2               | 280         | 117     | 520       | 112     |
| 3/16 in            |                        |                       |     |                   | 230         | 118     | 410       | 115     |
| 1/4 in             |                        |                       |     | 0.5               | 175         | 119     | 275       | 119     |
| 3/8 in             |                        |                       |     |                   | 125         | 120     | 200       | 120     |
| 1/2 in             |                        | 0.24                  | 300 | 0.6               | 65          | 122     | 108       | 122     |
| 5/8 in             |                        | Edge Start            |     |                   | 42          | 127     | 70        | 124     |
| 3/4 in             |                        |                       |     |                   | 33          | 131     | 46        | 126     |
| 7/8 in             |                        |                       |     |                   | 23          | 136     | 34        | 130     |
| 1 in               |                        |                       |     |                   | 18          | 140     | 26        | 134     |

## TORCH SETUP

### 85A Unshielded Stainless Steel

| Air flow rate - slpm/scfh |           |
|---------------------------|-----------|
| Hot                       | 190 / 400 |
| Cold                      | 235 / 500 |

#### Metric

| Material Thickness | Torch-to-Work Distance | Initial Pierce Height |     | Pierce Delay Time | Recommended |         | Maximum   |         |     |
|--------------------|------------------------|-----------------------|-----|-------------------|-------------|---------|-----------|---------|-----|
|                    |                        |                       |     |                   | Cut Speed   | Voltage | Cut Speed | Voltage |     |
| mm                 | mm                     | mm                    | %   | seconds           | (mm/min)    | Volts   | (mm/min)  | Volts   |     |
| 2                  | 2.0                    | 5.0                   | 250 | 0.1               | 8550        | 117     | 14000     | 114     |     |
| 3                  |                        |                       |     |                   | 7000        | 118     | 12000     | 116     |     |
| 4                  |                        |                       |     | 0.2               | 5600        | 118     | 9800      | 118     |     |
| 6                  |                        |                       |     |                   | 0.5         | 3400    | 120       | 5700    | 122 |
| 8                  |                        | 2250                  | 121 | 3700              |             | 124     |           |         |     |
| 10                 |                        | 6.0                   | 300 | 0.5               | 1430        | 123     | 2300      | 125     |     |
| 12                 |                        |                       |     | 0.7               | 1000        | 129     | 1700      | 127     |     |
| 16                 |                        | Edge Start            |     |                   |             | 650     | 134       | 910     | 131 |
| 20                 |                        |                       |     |                   |             | 360     | 138       | 720     | 136 |

#### English

| Material Thickness | Torch-to-Work Distance | Initial Pierce Height |     | Pierce Delay Time | Recommended |         | Maximum   |         |
|--------------------|------------------------|-----------------------|-----|-------------------|-------------|---------|-----------|---------|
|                    |                        |                       |     |                   | Cut Speed   | Voltage | Cut Speed | Voltage |
|                    | in                     | in                    | %   | seconds           | ipm         | Volts   | ipm       | Volts   |
| 14GA               | 0.08                   | 0.20                  | 250 | 0.1               | 365         | 117     | 580       | 114     |
| 10GA               |                        |                       |     | 0.2               | 280         | 120     | 460       | 116     |
| 3/16 in            |                        |                       |     |                   | 200         | 123     | 335       | 118     |
| 1/4 in             |                        |                       |     | 0.5               | 110         | 126     | 215       | 119     |
| 3/8 in             |                        |                       |     |                   | 75          | 127     | 110       | 123     |
| 1/2 in             |                        |                       |     | 0.24              | 300         | 0.6     | 45        | 135     |
| 5/8 in             |                        | Edge Start            |     |                   | 34          | 139     | 56        | 132     |
| 3/4 in             |                        |                       |     |                   | 22          | 143     | 40        | 136     |

## TORCH SETUP

### 85A Unshielded

#### Aluminum

| Air flow rate - slpm/scfh |           |
|---------------------------|-----------|
| Hot                       | 190 / 400 |
| Cold                      | 235 / 500 |

#### Metric

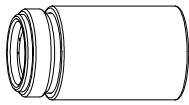
| Material Thickness | Torch-to-Work Distance | Initial Pierce Height |     | Pierce Delay Time | Recommended |         | Maximum   |         |     |
|--------------------|------------------------|-----------------------|-----|-------------------|-------------|---------|-----------|---------|-----|
|                    |                        |                       |     |                   | Cut Speed   | Voltage | Cut Speed | Voltage |     |
| mm                 | mm                     | mm                    | %   | seconds           | (mm/min)    | Volts   | (mm/min)  | Volts   |     |
| 2                  | 2.0                    | 5.0                   | 250 | 0.1               | 8700        | 118     | 14000     | 115     |     |
| 3                  |                        |                       |     |                   | 7350        | 120     | 12000     | 116     |     |
| 4                  |                        |                       |     | 0.2               | 6000        | 122     | 10000     | 117     |     |
| 6                  |                        |                       |     |                   | 0.5         | 3300    | 125       | 6150    | 119 |
| 8                  |                        |                       |     | 2350              |             | 127     | 4100      | 121     |     |
| 10                 |                        | 6.0                   | 300 | 0.5               | 1800        | 128     | 2650      | 124     |     |
| 12                 |                        |                       |     | 0.7               | 1300        | 133     | 2160      | 127     |     |
| 16                 |                        | Edge Start            |     |                   |             | 840     | 139       | 1400    | 132 |
| 20                 |                        |                       |     |                   |             | 470     | 144       | 900     | 137 |

#### English

| Material Thickness | Torch-to-Work Distance | Initial Pierce Height |     | Pierce Delay Time | Recommended |         | Maximum   |         |
|--------------------|------------------------|-----------------------|-----|-------------------|-------------|---------|-----------|---------|
|                    |                        |                       |     |                   | Cut Speed   | Voltage | Cut Speed | Voltage |
|                    | in                     | in                    | %   | seconds           | ipm         | Volts   | ipm       | Volts   |
| 1/8 in             | 0.08                   | 0.20                  | 250 | 0.2               | 280         | 120     | 460       | 116     |
| 3/16 in            |                        |                       |     |                   | 200         | 123     | 335       | 118     |
| 1/4 in             |                        |                       |     | 0.5               | 110         | 126     | 215       | 119     |
| 3/8 in             |                        |                       |     |                   | 75          | 127     | 110       | 123     |
| 1/2 in             |                        | 0.24                  | 300 | 0.6               | 45          | 135     | 78        | 128     |
| 5/8 in             |                        | Edge Start            |     |                   | 34          | 139     | 56        | 132     |
| 3/4 in             |                        |                       |     |                   | 22          | 143     | 40        | 136     |

**65 A unshielded consumables**

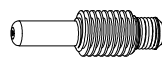
220955  
Deflector



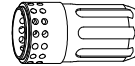
220854  
Retaining cap



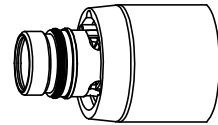
220819  
Nozzle



220842  
Electrode



220857  
Swirl ring



## TORCH SETUP

65A Unshielded

Mild Steel

| Air flow rate - slpm/scfh |           |
|---------------------------|-----------|
| Hot                       | 160 / 340 |
| Cold                      | 220 / 470 |

Metric

| Material Thickness | Torch-to-Work Distance | Initial Pierce Height |     | Pierce Delay Time | Recommended |         | Maximum   |         |
|--------------------|------------------------|-----------------------|-----|-------------------|-------------|---------|-----------|---------|
|                    |                        |                       |     |                   | Cut Speed   | Voltage | Cut Speed | Voltage |
| mm                 | mm                     | mm                    | %   | seconds           | (mm/min)    | Volts   | (mm/min)  | Volts   |
| 2                  | 2.0                    | 5.0                   | 250 | 0.1               | 6050        | 117     | 9200      | 116     |
| 3                  |                        |                       |     | 0.2               | 5200        | 118     | 7900      | 117     |
| 4                  |                        |                       |     | 0.5               | 4250        | 118     | 6550      | 117     |
| 6                  |                        |                       |     |                   | 2550        | 120     | 4100      | 119     |
| 8                  |                        |                       |     |                   | 1620        | 123     | 2800      | 120     |
| 10                 |                        | 6.0                   | 300 | 0.7               | 970         | 127     | 1880      | 122     |
| 12                 |                        | Edge Start            |     |                   | 760         | 129     | 1400      | 124     |
| 16                 |                        |                       |     |                   | 500         | 134     | 800       | 128     |
| 20                 |                        |                       |     |                   | 280         | 138     | 560       | 132     |

English

| Material Thickness | Torch-to-Work Distance | Initial Pierce Height |     | Pierce Delay Time | Recommended |         | Maximum   |         |
|--------------------|------------------------|-----------------------|-----|-------------------|-------------|---------|-----------|---------|
|                    |                        |                       |     |                   | Cut Speed   | Voltage | Cut Speed | Voltage |
|                    | in                     | in                    | %   | seconds           | ipm         | Volts   | ipm       | Volts   |
| 16GA               | 0.08                   | 0.20                  | 250 | 0.1               | 255         | 116     | 385       | 115     |
| 10GA               |                        |                       |     |                   | 190         | 118     | 290       | 117     |
| 3/16 in            |                        |                       |     | 0.2               | 135         | 119     | 215       | 118     |
| 1/4 in             |                        |                       |     | 0.5               | 90          | 120     | 145       | 119     |
| 3/8 in             |                        | 0.24                  | 300 | 0.7               | 40          | 126     | 78        | 121     |
| 1/2 in             |                        | Edge Start            |     |                   | 27          | 130     | 50        | 125     |
| 5/8 in             |                        |                       |     |                   | 20          | 134     | 32        | 128     |
| 3/4 in             |                        |                       |     |                   | 13          | 137     | 22        | 131     |

## TORCH SETUP

### 65A Unshielded Stainless Steel

| Air flow rate - slpm/scfh |           |
|---------------------------|-----------|
| Hot                       | 160 / 340 |
| Cold                      | 220 / 470 |

#### Metric

| Material Thickness | Torch-to-Work Distance | Initial Pierce Height |     | Pierce Delay Time | Recommended |         | Maximum   |         |
|--------------------|------------------------|-----------------------|-----|-------------------|-------------|---------|-----------|---------|
|                    |                        |                       |     |                   | Cut Speed   | Voltage | Cut Speed | Voltage |
| mm                 | mm                     | mm                    | %   | seconds           | (mm/min)    | Volts   | (mm/min)  | Volts   |
| 2                  | 2.0                    | 5.0                   | 250 | 0.1               | 7950        | 117     | 12800     | 119     |
| 3                  |                        |                       |     | 0.2               | 6600        | 118     | 10600     | 119     |
| 4                  |                        |                       |     | 0.5               | 5050        | 119     | 8200      | 119     |
| 6                  |                        |                       |     |                   | 2300        | 121     | 3800      | 120     |
| 8                  |                        |                       |     | 0.7               | 1400        | 123     | 2400      | 121     |
| 10                 |                        | 6.0                   | 300 | 0.7               | 920         | 126     | 1550      | 123     |
| 12                 |                        | Edge Start            |     |                   | 710         | 130     | 1150      | 127     |
| 16                 |                        |                       |     |                   | 480         | 135     | 630       | 132     |

#### English

| Material Thickness | Torch-to-Work Distance | Initial Pierce Height |     | Pierce Delay Time | Recommended |         | Maximum   |         |
|--------------------|------------------------|-----------------------|-----|-------------------|-------------|---------|-----------|---------|
|                    |                        |                       |     |                   | Cut Speed   | Voltage | Cut Speed | Voltage |
|                    | in                     | in                    | %   | seconds           | ipm         | Volts   | ipm       | Volts   |
| 16GA               | 0.08                   | 0.20                  | 250 | 0.1               | 340         | 116     | 540       | 118     |
| 10GA               |                        |                       |     |                   | 235         | 118     | 380       | 119     |
| 3/16 in            |                        |                       |     | 0.2               | 150         | 120     | 240       | 119     |
| 1/4 in             |                        |                       |     | 0.5               | 75          | 121     | 125       | 120     |
| 3/8 in             |                        | 0.24                  | 300 | 0.7               | 38          | 125     | 65        | 122     |
| 1/2 in             |                        | Edge Start            |     |                   | 25          | 132     | 40        | 128     |
| 5/8 in             |                        |                       |     |                   | 19          | 135     | 25        | 132     |

## TORCH SETUP

65A Unshielded

Aluminum

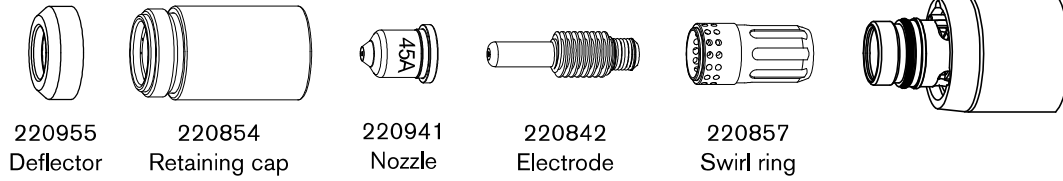
| Air flow rate - slpm/scfh |           |
|---------------------------|-----------|
| Hot                       | 160 / 340 |
| Cold                      | 220 / 470 |

Metric

| Material Thickness | Torch-to-Work Distance | Initial Pierce Height |     | Pierce Delay Time | Recommended |         | Maximum   |         |
|--------------------|------------------------|-----------------------|-----|-------------------|-------------|---------|-----------|---------|
|                    |                        |                       |     |                   | Cut Speed   | Voltage | Cut Speed | Voltage |
| mm                 | mm                     | mm                    | %   | seconds           | (mm/min)    | Volts   | (mm/min)  | Volts   |
| 2                  | 2.0                    | 5.0                   | 250 | 0.1               | 7750        | 123     | 14200     | 123     |
| 3                  |                        |                       |     | 0.2               | 6550        | 124     | 12000     | 123     |
| 4                  |                        |                       |     | 0.5               | 5400        | 125     | 9500      | 123     |
| 6                  |                        |                       |     |                   | 3000        | 127     | 4850      | 124     |
| 8                  |                        |                       |     | 0.7               | 1800        | 130     | 3000      | 126     |
| 10                 |                        | 6.0                   | 300 | 0.7               | 1100        | 133     | 2050      | 127     |
| 12                 |                        | Edge Start            |     |                   | 900         | 135     | 1560      | 129     |
| 16                 |                        |                       |     |                   | 600         | 139     | 880       | 132     |

English

| Material Thickness | Torch-to-Work Distance | Initial Pierce Height |     | Pierce Delay Time | Recommended |         | Maximum   |         |
|--------------------|------------------------|-----------------------|-----|-------------------|-------------|---------|-----------|---------|
|                    |                        |                       |     |                   | Cut Speed   | Voltage | Cut Speed | Voltage |
|                    | in                     | in                    | %   | seconds           | ipm         | Volts   | ipm       | Volts   |
| 1/16 in            | 0.08                   | 0.20                  | 250 | 0.1               | 325         | 122     | 590       | 122     |
| 1/8 in             |                        |                       |     |                   | 175         | 125     | 300       | 123     |
| 1/4 in             |                        |                       |     | 0.5               | 100         | 127     | 160       | 124     |
| 3/8 in             |                        | 0.24                  | 300 | 0.7               | 45          | 132     | 85        | 127     |
| 1/2 in             |                        | Edge Start            |     |                   | 32          | 136     | 55        | 129     |
| 5/8 in             |                        |                       |     |                   | 24          | 138     | 35        | 132     |

**45 A unshielded consumables**

## TORCH SETUP

45A Unshielded

Mild Steel

| Air flow rate - slpm/scfh |           |
|---------------------------|-----------|
| Hot                       | 147 / 310 |
| Cold                      | 210 / 450 |

Metric

| Material Thickness | Torch-to-Work Distance | Initial Pierce Height |     | Pierce Delay Time | Recommended |         | Maximum   |         |
|--------------------|------------------------|-----------------------|-----|-------------------|-------------|---------|-----------|---------|
|                    |                        |                       |     |                   | Cut Speed   | Voltage | Cut Speed | Voltage |
| mm                 | mm                     | mm                    | %   | seconds           | (mm/min)    | Volts   | (mm/min)  | Volts   |
| 0.5                | 1.5                    | 3.8                   | 250 | 0.0               | 9000        | 120     | 12700     | 122     |
| 1                  |                        |                       |     |                   | 9000        | 120     | 12700     | 122     |
| 1.5                |                        |                       |     | 0.1               | 7700        | 120     | 12700     | 122     |
| 2                  |                        |                       |     | 0.3               | 6150        | 119     | 9750      | 120     |
| 3                  |                        |                       |     | 0.4               | 3950        | 121     | 6150      | 122     |
| 4                  |                        |                       |     |                   | 2350        | 123     | 4450      | 124     |
| 6                  |                        |                       |     | 0.5               | 1400        | 126     | 2550      | 123     |

English

| Material Thickness | Torch-to-Work Distance | Initial Pierce Height |     | Pierce Delay Time | Recommended |         | Maximum   |         |
|--------------------|------------------------|-----------------------|-----|-------------------|-------------|---------|-----------|---------|
|                    |                        |                       |     |                   | Cut Speed   | Voltage | Cut Speed | Voltage |
|                    | in                     | in                    | %   | seconds           | ipm         | Volts   | ipm       | Volts   |
| 26GA               | 0.06                   | 0.15                  | 250 | 0.0               | 350         | 120     | 500       | 122     |
| 22GA               |                        |                       |     |                   | 350         | 120     | 500       | 122     |
| 18GA               |                        |                       |     | 0.1               | 350         | 119     | 500       | 122     |
| 16GA               |                        |                       |     |                   | 300         | 121     | 500       | 122     |
| 14GA               |                        |                       |     | 0.2               | 250         | 119     | 400       | 120     |
| 12GA               |                        |                       |     | 0.4               | 200         | 120     | 270       | 122     |
| 10GA               |                        |                       |     |                   | 100         | 123     | 205       | 123     |
| 3/16 in            |                        |                       |     | 0.5               | 85          | 122     | 135       | 125     |
| 1/4 in             |                        |                       |     | 0.6               | 48          | 127     | 91        | 122     |

## TORCH SETUP

### 45A Unshielded Stainless Steel

| Air flow rate - slpm/scfh |           |
|---------------------------|-----------|
| Hot                       | 147 / 310 |
| Cold                      | 210 / 450 |

#### Metric

| Material Thickness | Torch-to-Work Distance | Initial Pierce Height |     | Pierce Delay Time | Recommended |         | Maximum   |         |
|--------------------|------------------------|-----------------------|-----|-------------------|-------------|---------|-----------|---------|
|                    |                        |                       |     |                   | Cut Speed   | Voltage | Cut Speed | Voltage |
| mm                 | mm                     | mm                    | %   | seconds           | (mm/min)    | Volts   | (mm/min)  | Volts   |
| 0.5                | 1.5                    | 3.8                   | 250 | 0.0               | 9000        | 121     | 12700     | 119     |
| 1                  |                        |                       |     |                   | 9000        | 121     | 12700     | 119     |
| 1.5                |                        |                       |     | 0.1               | 9000        | 121     | 12700     | 119     |
| 2                  |                        |                       |     | 0.3               | 6000        | 122     | 12700     | 119     |
| 3                  |                        |                       |     | 0.4               | 3250        | 123     | 5950      | 119     |
| 4                  |                        |                       |     |                   | 1900        | 128     | 3700      | 119     |
| 6                  |                        |                       |     | 0.5               | 700         | 130     | 1800      | 127     |

#### English

| Material Thickness | Torch-to-Work Distance | Initial Pierce Height |     | Pierce Delay Time | Recommended |         | Maximum   |         |
|--------------------|------------------------|-----------------------|-----|-------------------|-------------|---------|-----------|---------|
|                    |                        |                       |     |                   | Cut Speed   | Voltage | Cut Speed | Voltage |
|                    | in                     | in                    | %   | seconds           | ipm         | Volts   | ipm       | Volts   |
| 26GA               | 0.06                   | 0.15                  | 250 | 0.0               | 350         | 120     | 500       | 119     |
| 22GA               |                        |                       |     |                   | 350         | 120     | 500       | 119     |
| 18GA               |                        |                       |     | 0.1               | 350         | 118     | 500       | 119     |
| 16GA               |                        |                       |     |                   | 350         | 121     | 500       | 119     |
| 14GA               |                        |                       |     | 0.2               | 300         | 122     | 500       | 117     |
| 12GA               |                        |                       |     | 0.4               | 150         | 121     | 280       | 119     |
| 10GA               |                        |                       |     |                   | 100         | 125     | 175       | 119     |
| 3/16 in            |                        |                       |     | 0.5               | 42          | 131     | 110       | 120     |
| 1/4 in             |                        |                       |     | 0.6               | 25          | 130     | 60        | 129     |

## TORCH SETUP

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**45A Unshielded**

**Aluminum**

| Air flow rate - slpm/scfh |           |
|---------------------------|-----------|
| Hot                       | 147 / 310 |
| Cold                      | 210 / 450 |

**Metric**

| Material Thickness | Torch-to-Work Distance | Initial Pierce Height |     | Pierce Delay Time | Recommended |         | Maximum   |         |
|--------------------|------------------------|-----------------------|-----|-------------------|-------------|---------|-----------|---------|
|                    |                        |                       |     |                   | Cut Speed   | Voltage | Cut Speed | Voltage |
| mm                 | mm                     | mm                    | %   | seconds           | (mm/min)    | Volts   | (mm/min)  | Volts   |
| 1                  | 1.5                    | 3.8                   | 250 | 0.0               | 7400        | 126     | 12700     | 119     |
| 2                  |                        |                       |     | 0.1               | 4400        | 127     | 11500     | 119     |
| 3                  |                        |                       |     | 0.2               | 2800        | 129     | 7800      | 122     |
| 4                  |                        |                       |     | 0.4               | 2100        | 132     | 5850      | 123     |
| 6                  |                        |                       |     | 0.5               | 1050        | 135     | 2800      | 126     |

**English**

| Material Thickness | Torch-to-Work Distance | Initial Pierce Height |     | Pierce Delay Time | Recommended |         | Maximum   |         |
|--------------------|------------------------|-----------------------|-----|-------------------|-------------|---------|-----------|---------|
|                    |                        |                       |     |                   | Cut Speed   | Voltage | Cut Speed | Voltage |
|                    | in                     | in                    | %   | seconds           | ipm         | Volts   | ipm       | Volts   |
| 1/32 in            | 0.06                   | 0.15                  | 250 | 0.0               | 325         | 126     | 500       | 120     |
| 1/16 in            |                        |                       |     | 0.1               | 200         | 126     | 500       | 116     |
| 3/32 in            |                        |                       |     | 0.2               | 150         | 127     | 410       | 122     |
| 1/8 in             |                        |                       |     | 0.4               | 100         | 130     | 280       | 122     |
| 1/4 in             |                        |                       |     | 0.5               | 36          | 136     | 90        | 126     |